Database Management Systems (DBMS)

Project Report Format

Inventory Management System

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General Instructions

1. The front end may be developed using any suitable programming language or framework.
2. **MySQL is mandatory** for database creation, storage, and query execution. A different DBMS must not be used in place of MySQL.
3. Design the database before implementing the application.
4. Demonstrate the DBMS concepts covered in the course through the report and implementation.
5. Include all diagrams, functional dependencies, normalization steps, SQL statements, and execution results.
6. Submit the report, application source code, and a complete MySQL script or database export.
7. Every team member must understand the database design and be prepared to explain it during evaluation.

Important: The project title used throughout this document is an example. Students must replace it with their own approved project topic and adapt the entities, business rules, relations, and queries accordingly.

Chapter 1

Introduction

1.1 Project Overview

Introduce the project and explain its purpose and intended users.

1.2 Problem Statement

Describe the problem the project aims to solve and the limitations of the existing approach.

1.3 Objectives

List the main objectives of the project.

1.4 Scope of the Project

Explain the functionalities covered by the project and its intended users.

1.5 Software and Hardware Requirements

Mention the programming language, front-end framework, MySQL version, development tools, and minimum hardware requirements.

Chapter 2

System Requirements and Analysis

2.1 Functional Requirements

List the operations the system must support. For an Inventory Management System, examples include:

- Adding, updating, searching, and deleting product records.
- Managing suppliers and product categories.
- Recording purchases and sales.
- Updating stock quantities.
- Generating low-stock alerts and inventory reports.

2.2 Business Rules

State the rules governing the application. Examples include:

- Each product must have a unique product ID.
- Each supplier must have a unique supplier ID.
- Each sale must contain at least one product.
- Stock quantity must not become negative.
- Each purchase or sale must be associated with a valid transaction record.

Chapter 3

Entity–Relationship (ER) Modeling

3.1 Identification of Entities

Identify the entities required for the project and list their attributes.
Example entities: Product, Supplier, Category, Purchase, Sale, and Sale_Details.

3.2 Identification of Relationships

Identify the relationships among entities and specify their cardinalities.
Examples: A category contains multiple products; a supplier can supply multiple products; a sale can contain multiple products.

3.3 ER Diagram

Insert the complete ER diagram. It must clearly show:

- Entities and their attributes.
- Primary keys.
- Relationships and relationship attributes, where applicable.
- Cardinality ratios (1:1, 1:N, or M:N).
- Total or partial participation constraints, wherever applicable.

3.4 Explanation of the ER Diagram

Explain the entities, relationships, cardinalities, and participation constraints used in the diagram.

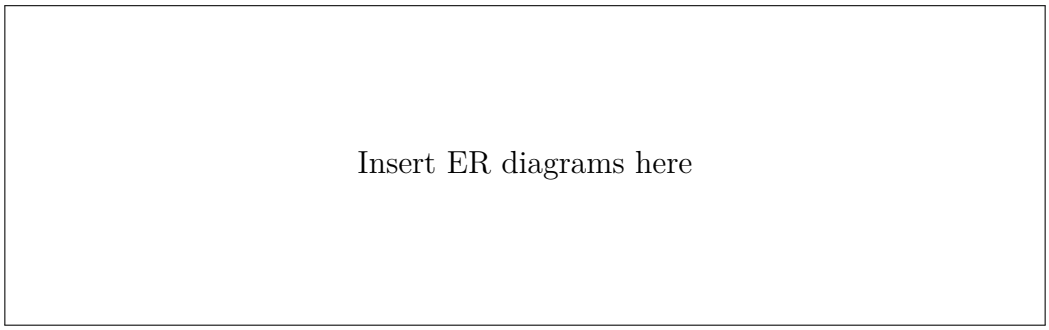


Figure 3.1: ER diagram(s) of the proposed system

Chapter 4

ER-to-Relational Mapping

4.1 Mapping Rules

Briefly state the mapping rules used to convert the ER diagram into relational tables.

4.2 Step-by-Step Conversion

Show how each entity and relationship is converted into a relation. Include the conversion of:

- Strong entities.
- Weak entities, if applicable.
- 1:1 relationships.
- 1:N relationships.
- M:N relationships.
- Multivalued attributes, if applicable.
- Relationships with their own attributes, if applicable.

4.3 Final Relational Schema

Write the complete relational schema. For example:

```
PRODUCT(Product_ID, Product_Name, Price, Quantity, Category_ID)
```

Underline primary keys and clearly identify foreign keys. Also identify candidate keys and composite keys, wherever applicable.

Chapter 5

Functional Dependencies

5.1 Identification of Functional Dependencies

Identify the functional dependencies for each relation based on the business rules.
Example:

$$Product_ID \rightarrow Product_Name, Price, Quantity, Category_ID$$

5.2 Candidate Keys and Attribute Classification

Determine the candidate keys, prime attributes, and non-prime attributes of each relation.

5.3 FD Analysis

Explain the reasoning behind the identified functional dependencies. Each dependency must be justified by the application requirements; do not list dependencies arbitrarily.

Chapter 6

Database Normalization

Normalization must be demonstrated step by step, not merely stated.

6.1 Unnormalized Form (UNF)

Present an unnormalized relation, if applicable, and identify repeating groups or other problems.

6.2 First Normal Form (1NF)

Convert the relation into 1NF by ensuring that attribute values are atomic and repeating groups are eliminated.

6.3 Second Normal Form (2NF)

Identify partial functional dependencies and remove them where applicable.

6.4 Third Normal Form (3NF)

Identify transitive dependencies and decompose relations where required.

6.5 Boyce–Codd Normal Form (BCNF)

Check whether each resulting relation satisfies BCNF. If a relation violates BCNF, demonstrate the decomposition. If no violation exists, explain why the relation satisfies BCNF.

6.6 Final Normalized Schema

Present the final set of relations after normalization. Show intermediate relations and explain each decomposition. Merely stating that the database is in 3NF or BCNF is insufficient.

Chapter 7

Relational Integrity and Constraints

Explain the integrity constraints applied to the database, including:

- Domain constraints.
- Primary key constraints.
- Foreign key constraints.
- `UNIQUE` constraints.
- `NOT NULL` constraints.
- `CHECK` constraints, wherever applicable.
- Referential actions such as `ON DELETE` or `ON UPDATE`, wherever appropriate.

  Explain how these constraints maintain data consistency and prevent invalid data.

Chapter 8

MySQL Database Implementation

8.1 Database Creation

Provide the SQL commands used to create the database.

8.2 Table Creation

Provide the complete `CREATE TABLE` statements for all relations, including primary keys, foreign keys, and other constraints.

8.3 Data Insertion

Provide sample `INSERT` statements and insert sufficient meaningful data to demonstrate the system.

8.4 Data Modification and Deletion

Demonstrate appropriate `UPDATE` and `DELETE` operations.

8.5 Sample Database Contents

Show table contents using `SELECT` queries or screenshots from MySQL.

8.6 SQL Query Implementation

Write and execute queries demonstrating the following concepts:

1. Simple `SELECT` queries with conditions.
2. Sorting using `ORDER BY`.

3. Aggregate functions: `COUNT`, `SUM`, `AVG`, `MIN`, and `MAX`.
4. Grouping using `GROUP BY` and filtering groups using `HAVING`.
5. Joins, including `INNER JOIN` and `LEFT JOIN`, along with other relevant joins where appropriate.
6. Nested queries and subqueries.
7. Queries using `IN`, `EXISTS`, `ANY`, or `ALL`, wherever relevant.
8. Queries involving multiple tables.
9. Queries that answer meaningful business questions.
10. At least one view, where appropriate.

For every query, provide the SQL statement, its purpose, and the output or a screenshot of its execution.

8.7 Advanced DBMS Features

Demonstrate at least two relevant features from the following list, preferably those covered in the course:

- Views.
- Triggers.

Explain the purpose and working of each implemented feature.

Chapter 9

Application Development

9.1 System Architecture

Show the interaction among the front end, application logic, and MySQL database using a simple architecture diagram.

9.2 Front-End Technologies

Mention the programming language, libraries, and frameworks used.

9.3 Application Modules

Explain the modules implemented in the project. For an Inventory Management System, examples include:

- Product Management.
- Supplier Management.
- Purchase Management.
- Sales Management.
- Stock Management.
- Reports and Alerts.

9.4 Database Connectivity

Explain how the application connects to MySQL and executes database operations.

9.5 Application Screenshots

Include screenshots of the major application screens and their corresponding operations.

9.6 Database Operations through the Application

Demonstrate that records can be inserted, retrieved, updated, and deleted through the application wherever appropriate. Show that changes are reflected in the MySQL database.

Chapter 10

Testing and Results

10.1 Test Cases

Prepare a table containing the test case ID, operation, input, expected result, actual result, and status.

Test ID	**Operation**	**Input**	**Expected Result**	**Actual Result**	**Status**
T1	Add product	Valid product data	Product added	Fill in result	Pass/Fail
T2	Duplicate ID	Existing product ID	Rejected by constraint	Fill in result	Pass/Fail
T3	Invalid foreign key	Non-existent category ID	Rejected by constraint	Fill in result	Pass/Fail
T4	Negative stock	Negative quantity	Rejected, if constrained	Fill in result	Pass/Fail

10.2 Query Testing

Show that the SQL queries produce the expected results.

10.3 Constraint Testing

Demonstrate that the database rejects invalid operations wherever the relevant constraints apply.

10.4 Results and Discussion

Summarize the functionality implemented and discuss the results obtained.

Chapter 11

Conclusion and Future Scope

11.1 Conclusion

Summarize the project, the database concepts applied, and the main outcomes.

11.2 Limitations

Mention the limitations of the current implementation.

11.3 Future Scope

Suggest possible improvements and additional functionalities.

References

List the books, official documentation, websites, and other resources consulted during the project. Use one consistent referencing style.

Appendix A

Complete MySQL Script

Include the database creation commands, table definitions, constraints, sample data, SQL queries, and any views, triggers, procedures, or transactions implemented. Alternatively, provide a clearly identified supplementary SQL file with the report.

Appendix B

Source Code

Provide the source code or a repository link for the application. Ensure that the submitted code is accessible to the evaluator.

Appendix C

Additional Screenshots

Include any additional screenshots needed to demonstrate the project.

Appendix D

Team Contribution

For group projects, clearly mention the contribution of each team member.

Suggested Evaluation Scheme

Component	Marks
Problem analysis and requirements	5
ER diagram and constraints	10
ER-to-relational mapping	10
Functional dependencies and keys	10
Normalization	15
MySQL schema, constraints, and sample data	10
SQL queries and advanced DBMS features	15
Front-end integration and functionality	10
Testing and report quality	5
Viva and conceptual understanding	10
Total	**100**

Evaluation principle: Marks should primarily reflect the correctness of database design, ER-to-relational mapping, functional dependencies, normalization, SQL implementation, and understanding of DBMS concepts. Front-end appearance alone must not determine the project grade.